

DESIGN TASK SIMULATION

Title

Cloud-Based Radio Resource Allocation Using MATLAB

Aim

To design and simulate a cloud-based radio resource allocation model that allocates available Resource Blocks (RBs) to multiple users based on their resource demand and priority.

Objective

- To generate different resource demands for users.
- To assign priority levels to users.
- To allocate available Resource Blocks based on priority.
- To analyze the difference between requested and allocated resources.
- To evaluate resource utilization.

Simulation Model

Input → Cloud Resource Manager → Priority Sorting → Resource Allocation → Output

Input:

- Number of users = 10
- Total Resource Blocks = 100
- User demand = 5–20 RBs
- Priority = 1–3

Processing:

1. Generate user demand and priority.
2. Sort users according to priority.
3. Allocate RBs to high-priority users first.
4. Update the remaining RBs.
5. Continue until resources are exhausted.

Output:

- User demand
- User priority
- Allocated RBs
- Demand vs Allocation graph
- Resource utilization pie chart

Simulation code:

```
clc;
```

```
clear;
```

```
close all;
```

```
numUsers = 10;
```

```
totalRB = 100;
```

```
demand = randi([5 20],1,numUsers);
```

```
priority = randi([1 3],1,numUsers);
```

```
allocation = zeros(1,numUsers);
```

```
remainingRB = totalRB;
```

```
[~,index] = sort(priority,'descend');
```

```
for i = 1:numUsers
```

```
    u = index(i);
```

```
    if remainingRB >= demand(u)
```

```
        allocation(u) = demand(u);
```

```
else
    allocation(u) = remainingRB;
end

remainingRB = remainingRB - allocation(u);

if remainingRB == 0
    break;
end
end

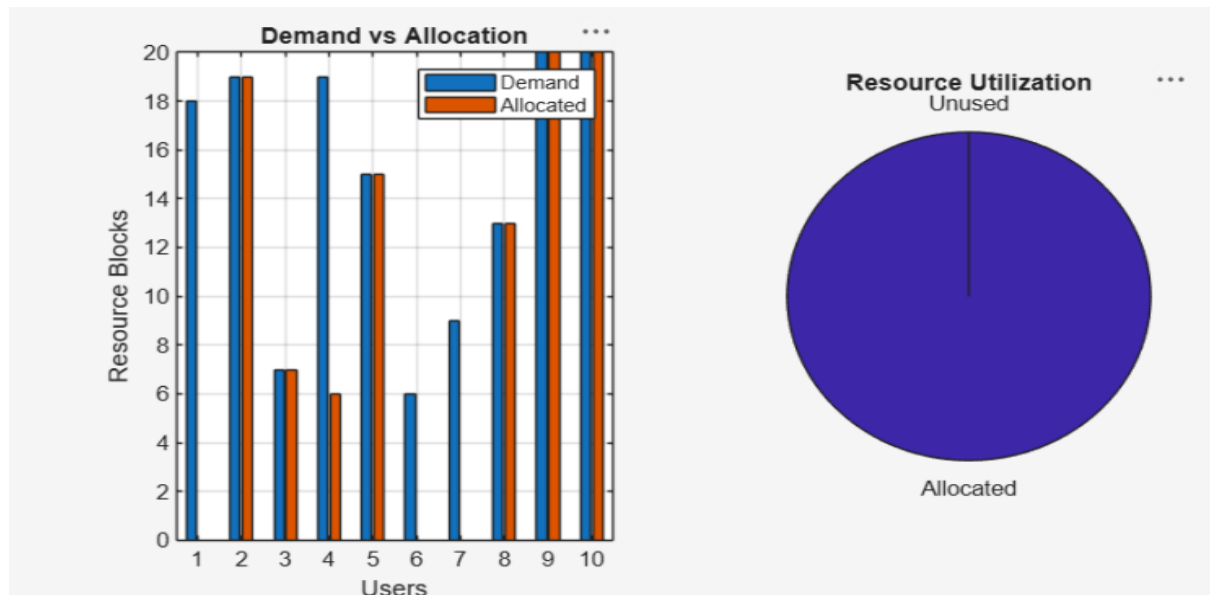
disp('User Demand Priority Allocation');
disp([(1:numUsers)' demand' priority' allocation']);

figure;

subplot(1,2,1);
bar([demand' allocation'],'grouped');
xlabel('Users');
ylabel('Resource Blocks');
title('Demand vs Allocation');
legend('Demand','Allocated');
grid on;

subplot(1,2,2);
pie([sum(allocation) remainingRB],{'Allocated','Unused'});
title('Resource Utilization');
```

Output:



Simulation Graphs:

Graph 1: Demand vs Allocation

- Compares the resources requested by each user with the resources actually allocated.
- High-priority users are served first.

Graph 2: Resource Utilization

- Shows the percentage of RBs that are **allocated** and **unused**.

Result

The proposed **cloud-based radio resource allocation model** was successfully designed and simulated using MATLAB. The available Resource Blocks were dynamically allocated to users according to their priority and demand. The simulation demonstrates efficient utilization of cloud radio resources and provides a simple model for priority-based resource management.

